Computer Networks: Routing, Multicasting and IPv6

Forwarding, Routing Algorithms, Protocols and Addressing

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Classroom Lecture

The Network Layer: Big Picture

Main responsibility

The network layer delivers packets from a source host to a destination host across one or more interconnected networks.

Forwarding

- Local, per-router action
- Chooses the outgoing interface for a packet
- Uses the forwarding table

Routing

- Network-wide process
- Learns or calculates paths
- Populates routing information

Routing decides the paths; forwarding moves each packet to the next hop.

Forwarding of IP Packets

When a router receives an IP packet, it typically:

1. Checks the packet's destination IP address.
2. Examines its forwarding/routing information.
3. Finds the best matching destination prefix.
4. Determines the next hop and outgoing interface.
5. Decreases the IPv4 TTL (or IPv6 Hop Limit).
6. Re-encapsulates the packet in a link-layer frame and sends it.

Important

The IP destination normally remains the final destination. The link-layer source and destination addresses change at each routed hop.

Longest-Prefix Match

Routers may have multiple routes that match a destination. The router generally selects the route with the **longest matching prefix**.

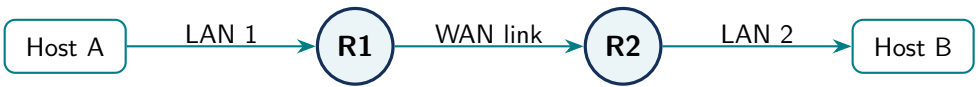
Destination prefix	Example match?	Prefix length
10.0.0.0/8	Yes	8 bits
10.1.0.0/16	Yes	16 bits
10.1.2.0/24	Yes	24 bits

For destination `10.1.2.55`, all three prefixes match. The router chooses `10.1.2.0/24`, because /24 is the most specific match.

Rule

More matching prefix bits = a more specific route.

Example: A Router Forwarding a Packet



- Host A sends a packet to Host B's IP address.
- R1 checks its route for Host B's destination network.
- R1 forwards the packet to R2, its next hop.
- R2 forwards it onto LAN 2 toward Host B.

Routing Table: What Does It Contain?

A routing table commonly records:

- Destination network or prefix
- Prefix length / subnet mask
- Next-hop IP address, if applicable
- Outgoing interface
- Metric or route preference
- Route source (connected, static, RIP, OSPF, etc.)

Destination	Next hop	Interface	Meaning
192.168.1.0/24	Direct	G0/0	Connected LAN
10.0.12.0/30	Direct	G0/1	Router link
192.168.2.0/24	10.0.12.2	G0/1	Remote LAN
0.0.0.0/0	10.0.12.2	G0/1	Default route

Static Routing

Definition

A static route is manually configured by an administrator. It does not automatically learn a replacement path from a routing protocol.

Advantages

- Simple and predictable for small networks.
- No routing-protocol update traffic.
- Useful for stub networks and controlled paths.

Limitations

- Manual configuration and maintenance.
- Does not automatically adapt to topology changes.
- Can become difficult to manage at scale.

Static Route: Cisco IOS Example

Suppose R1 must reach network `192.168.2.0/24` through next-hop `10.0.12.2`.

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.12.2
```

General syntax:

```
ip route <destination-network> <subnet-mask> <next-hop-IP>
```

Verify:

```
R1# show ip route
R1# show running-config
R1# ping 192.168.2.1
```

Check before testing

The interfaces must be up, IP addresses must be correct, and the return path must also exist.

Default Routing

Definition

A default route is used when no more-specific route matches the packet's destination.

IPv4 default prefix: `0.0.0.0/0`

- Often points toward an ISP or an upstream router.
- Avoids configuring a separate route for every unknown destination.
- A more-specific route is preferred over the default route.

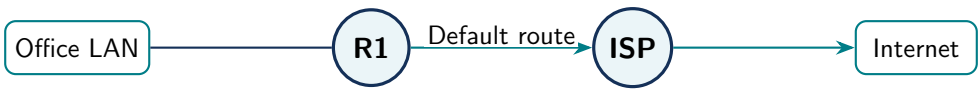
Example

A small office router knows its local LANs and sends all other traffic to the ISP router.

Default Route: Cisco IOS Example

```
R1(config)# ip route 0.0.0.0 0.0.0.0 10.0.12.2
```

Read it as: “For any IPv4 destination not covered by a better route, forward to 10.0.12.2.”



Static, Default and Dynamic Routing

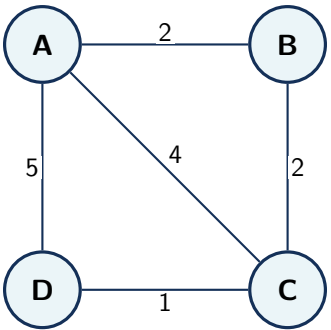
Type	How routes are learned	Typical use
Static	Entered manually	Small, stable or special-purpose networks
Default	Manually defines a fallback path; may also be distributed by a protocol	Stub networks, Internet gateway
Dynamic	Routers exchange information using protocols	Medium and large networks with changing topology

Key point

A default route is a route entry, not a separate dynamic routing algorithm. It can be configured statically or learned through a routing protocol.

Unicast Routing Algorithms

Unicast means one sender communicates with one destination. Routing algorithms determine which path should be used to reach a destination network.



The numbers represent illustrative link costs. An algorithm can compare possible paths and select one according to its rules and metric.

Three Main Routing Approaches

Approach	What routers learn	Common example
Distance Vector	Distance/metric and next hop learned from neighbours	RIP; EIGRP is an advanced distance-vector protocol
Link State	Network topology and link costs, then calculate shortest paths	OSPF
Path Vector	Reachability plus the path of autonomous systems and policy attributes	BGP

Memory aid

Distance Vector: "How far, through whom?"
Link State: "What is the topology?"

Distance Vector Routing: Core Idea

Each router maintains a table containing its best-known distance to destinations and the next hop to use.

- Routers exchange routing information with their neighbours.
- Each router uses received information to improve its own table.
- Updates are based on the distance advertised by a neighbour plus the cost to that neighbour.

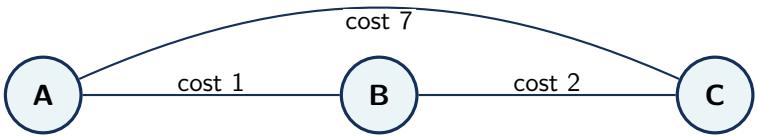
A simplified Bellman–Ford relation is:

$$D_x(y) = \min_{v \in N(x)} \{c(x, v) + D_v(y)\}$$

where:

- $D_x(y) =$ router x's current best distance to destination y
- $c(x, v) =$ cost from router x to neighbour v
- $D_v(y) =$ neighbour v's advertised distance to y

Distance Vector: Three-Router Example



- Direct A-to-C cost $= 7$.
- A learns from B that B can reach C at cost 2.
- A calculates the route through B: $1 + 2 = 3$.
- A replaces cost 7 with cost 3, using B as the next hop.

Lesson

Routers improve their routes by combining neighbour information with their own link costs.

Distance Vector: Simplified Routing Table

After convergence in the previous example:

Router A destination	Cost	Next hop	Reason
A (self)	0	–	Local router
B	1	B	Direct link
C	3	B	A–B–C is cheaper than direct cost 7

Convergence is the process by which routers' routing information settles after a change or update.

Remember

The metric is not always physical distance. It may represent hop count, cost, delay, bandwidth, or a combination, depending on the protocol.

Distance Vector: Strengths and Challenges

Strengths

- Conceptually simple.
- Routers need information from neighbours rather than a complete topology map.
- Suitable for smaller or simpler routing domains.

Challenges

- Convergence may be slow after failures.
- Incorrect information can circulate between routers.
- Some distance-vector protocols can experience the count-to-infinity problem.

Common mitigation techniques include split horizon, route poisoning, triggered updates and hold-down mechanisms, depending on the protocol.

RIP v1 and RIP v2

RIP (Routing Information Protocol) is a distance-vector interior gateway protocol.

Feature	RIP v1	RIP v2
Routing updates	Classful	Classless
Subnet mask in updates	Not carried	Carried
VLSM/CIDR support	No	Yes
Update delivery	Broadcast	Multicast to 224.0.0.9
Authentication support	Limited/not in standard v1	Supports authentication options
Metric	Hop count	Hop count

RIP metric

The maximum usable distance is 15 hops. A metric of 16 means the destination is unreachable.

RIP v2: Basic Cisco IOS Configuration

Example for a router with interfaces in networks `192.168.1.0/24` and `10.0.12.0/30`.

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
```

Verification:

```
Router# show ip route
Router# show ip protocols
Router# show ip rip database
```

Configuration note

The RIP `network` command identifies participating classful major networks in Cisco IOS; verify the exact commands supported by your device and software version.

Link State Routing: Core Idea

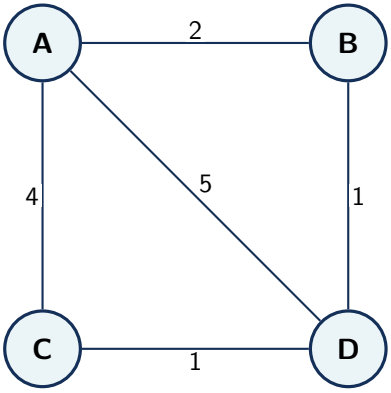
In link-state routing, routers build a shared view of the network topology and calculate shortest paths.

1. Discover directly connected neighbours.
2. Measure or assign the cost of each link.
3. Create link-state information describing local links.
4. Flood this information through the routing area.
5. Build a Link State Database (LSDB).
6. Run Dijkstra's shortest-path algorithm.

Key idea

Each router in the same area can independently calculate a shortest-path tree from its own point of view.

Link State: Topology Example



If router A calculates the path to D:

- A–B–D costs $2 + 1 = 3$.
- A–C–D costs $4 + 1 = 5$.
- Direct A–D costs 5.

The least-cost path is A–B–D, with total cost 3.

Dijkstra's Shortest-Path Algorithm

Dijkstra's algorithm repeatedly selects the not-yet-finalized node with the smallest tentative cost.

1. Set the source router's distance to 0; other distances start at infinity.
2. Select the unvisited node with the lowest tentative distance.
3. Update the distances to its neighbours if a cheaper path is found.
4. Mark that node as finalized.
5. Repeat until all reachable nodes are finalized.

Example from router A

Initial: $D(A) = 0$. From A, set $D(B) = 2$, $D(C) = 4$, $D(D) = 5$. After selecting B, update D to $2 + 1 = 3$. The best path to D is through B.

OSPF: Open Shortest Path First

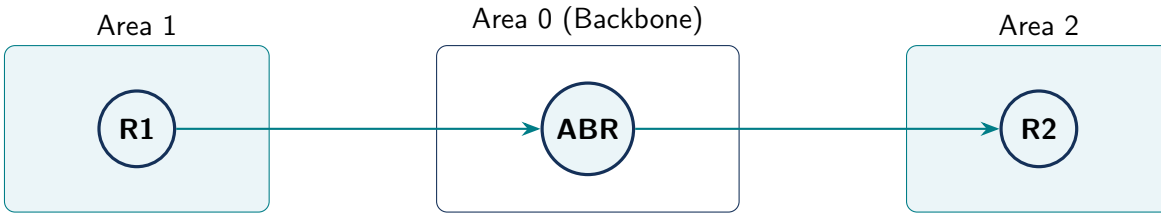
OSPF is a link-state Interior Gateway Protocol (IGP).

- Uses link-state information and Dijkstra's SPF algorithm.
- Metric is called cost; implementations commonly derive it from interface bandwidth.
- Routers exchange Hello packets to discover and maintain neighbours.
- LSAs describe topology information; the LSDB stores the area's link-state data.
- Supports hierarchical design using areas.

Area 0

Area 0 is the OSPF backbone. In a conventional multi-area design, other areas connect to the backbone through Area Border Routers (ABRs).

OSPF Areas: Conceptual View



- Areas limit the scope of some link-state information.
- An ABR connects an area to the OSPF backbone and exchanges routing information between areas.
- Multi-area design can improve scalability and control routing information.

OSPF: Basic Cisco IOS Configuration

Illustrative single-area configuration:

```
Router(config)# router ospf 1
Router(config-router)# router-id 1.1.1.1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.12.0 0.0.0.3 area 0
```

Useful verification commands:

```
Router# show ip ospf neighbor
Router# show ip ospf interface brief
Router# show ip route ospf
Router# show ip protocols
```

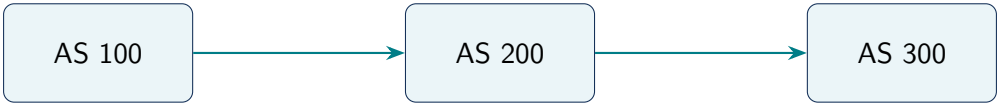
Common issue

OSPF neighbours require compatible settings and working IP connectivity on the shared link. Check area, timers, authentication, subnet, and interface status when adjacency fails.

Path Vector Routing

Path-vector routing advertises reachability together with information about the path to a destination.

- Used between autonomous systems (ASes).
- A route advertisement can include the sequence of AS numbers it traverses.
- Policies influence which routes are accepted and preferred.
- An AS can reject a route containing its own AS number, helping prevent routing loops.



Advertised AS path: 100 200

BGP: Border Gateway Protocol

BGP is the principal inter-domain routing protocol used to exchange reachability information between autonomous systems on the Internet.

- Uses TCP port 179 for peer sessions.
- Supports policy-based route selection rather than simply choosing the numerically shortest path.
- Uses attributes such as AS-PATH, NEXT-HOP, LOCAL-PREF and MED.
- eBGP exchanges routes between different ASes.
- iBGP distributes BGP routes within the same AS.

Important distinction

BGP's selected route is not necessarily the path with the fewest routers or lowest delay. Administrative policy and BGP attributes matter.

BGP: Conceptual Configuration

Illustrative Cisco IOS-style example (values must match the actual topology):

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 203.0.113.2 remote-as 65002
Router(config-router)# network 192.0.2.0 mask 255.255.255.0
```

- AS 65001 is the local autonomous system in this example.
- The neighbour is in AS 65002.
- The `network` statement advertises a prefix only when the corresponding route exists in the routing table, subject to platform/configuration requirements.

Lab note

The addresses shown are documentation/example ranges. Use addresses and AS numbers appropriate to your lab.

EIGRP: Enhanced Interior Gateway Routing Protocol

EIGRP is commonly described as an advanced distance-vector IGP.

- Uses the Diffusing Update Algorithm (DUAL).
- Maintains neighbour and topology information.
- Can identify a loop-free backup route called a feasible successor when conditions permit.
- Its composite metric commonly uses bandwidth and delay; reliability and load can also be configured.
- Supports partial and bounded updates rather than periodically sending the entire routing table in the same way as classic RIP.

Terminology

The **successor** is the selected best next hop. A **feasible successor** is a qualifying loop-free backup path.

EIGRP: Basic Cisco IOS Example

Illustrative classic-style configuration:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.12.0 0.0.0.3
```

Verification commands may include:

```
Router# show ip eigrp neighbors
Router# show ip eigrp topology
Router# show ip route eigrp
```

Version note

EIGRP configuration syntax and availability depend on IOS/IOS XE version and Packet Tracer device support. Some platforms use named configuration mode.

Compare the Main Routing Protocols

Protocol	Approach	Metric / selection	Typical scope	Key feature
RIP v1	Distance vector	Hop count	Inside an AS	Classful; 15-hop maximum
RIP v2	Distance vector	Hop count	Inside an AS	Classless; carries mask
OSPF	Link state	Cost	Inside an AS	SPF calculation; areas
EIGRP	Advanced distance vector	Composite metric	Inside an AS	DUAL; feasible successors
BGP	Path vector	Policy and attributes	Between ASes; also within an AS	Internet-scale policy routing

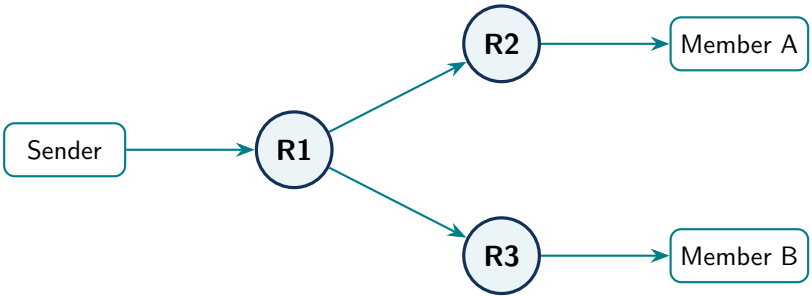
Exam tip

Do not confuse an algorithm's category with a protocol's metric. For example, RIP is distance vector and uses hop count; OSPF is link state and uses cost.

Unicast, Broadcast and Multicast

Delivery type	Communication pattern	Example
Unicast	One sender to one receiver	Web request to a server
Broadcast	One sender to all hosts in a broadcast domain (IPv4)	ARP request on a LAN
Multicast	One sender to a subscribed group of receivers	Live stream delivered to a group
Anycast	One address assigned at multiple locations; traffic goes to one selected instance	Some distributed services

Multicast: Basic Operation



- Receivers join a multicast group.
- Routers may build multicast distribution trees.
- Traffic is forwarded only along links that lead to group members, where multicast routing is configured.
- IGMP is used by IPv4 hosts to communicate group membership to local routers.

IPv4 Multicast Addressing

IPv4 multicast addresses are in the range:

`224.0.0.0` to `239.255.255.255`

This is the Class D range in traditional IPv4 classful terminology.

- A multicast IP address identifies a group, not one individual host interface.
- Hosts can join and leave multicast groups.
- Ethernet maps IPv4 multicast to corresponding multicast MAC addresses.
- Not every multicast address is intended for unrestricted Internet-wide use; many ranges have specific scopes or purposes.

Do not confuse

An IPv4 multicast group address is different from a broadcast address such as `255.255.255.255`.

Why IPv6?

IPv6 was developed primarily to address the exhaustion of the IPv4 address space and to support continued Internet growth.

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Representation	Decimal octets	Hexadecimal groups
Example	192.0.2.10	2001:db8::10
Broadcast	Supported	No broadcast; multicast is used
Header	Variable length	Fixed base header of 40 bytes
Address capacity	About 4.3 billion values	2^{128} possible values

IPv6 Address Structure and Notation

An IPv6 address contains 128 bits, conventionally written as eight groups of four hexadecimal digits separated by colons.

Expanded address

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Each group contains 16 bits. Leading zeros in any group may be omitted.

Shortened address

```
2001:db8:0:0:0:ff00:42:8329
```

A consecutive sequence of zero groups can be compressed once using `::`.

Compressed address

```
2001:db8::ff00:42:8329
```

IPv6 Zero Compression: Rules

1. Remove leading zeros within each 16-bit group.
2. Replace one consecutive run of all-zero groups with `::`.
3. Use `::` at most once in an address, because otherwise the number of omitted groups would be ambiguous.

Expanded	Compressed
2001:0db8:0000:0000:0000:0000:0000:0001	2001:db8::1
fe80:0000:0000:0000:0211:22ff:fe33:4455	fe80::211:22ff:fe33:4455
2001:0db8:0000:0000:0000:0000:0000:0000	2001:db8::

Important IPv6 Address Types

Type	Common prefix / pattern	Purpose
Global unicast	Commonly 2000::/3	Globally routable unicast addresses
Link-local unicast	fe80::/10	Communication on the local link; not forwarded by routers
Unique local	fc00::/7 (commonly fd00::/8 locally assigned)	Private-like internal addressing
Multicast	ff00::/8	One-to-many group delivery
Anycast	Uses unicast address format	Delivered to one selected member of an anycast group
Loopback	::1/128	A node refers to itself
Unspecified	::/128	Indicates no specified address; not assigned as a normal host address

IPv6 Prefix Length

IPv6 uses prefix notation similar to CIDR in IPv4.

Example

`2001:db8:1234:1::/64`

- `/64` means the first 64 bits are the network prefix.
- The remaining 64 bits identify the interface portion in a conventional /64 subnet.
- The prefix length is not the number of hexadecimal groups; it counts bits.

Practice

For `2001:db8:abcd:12::5/64`, the first four 16-bit groups form the /64 prefix: `2001:db8:abcd:12::/64`.

IPv6 Configuration: Basic Cisco IOS Example

```
Router(config)# ipv6 unicast-routing
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ipv6 address 2001:db8:1::1/64
Router(config-if)# no shutdown
```

Verification:

```
Router# show ipv6 interface brief
Router# show ipv6 route
Router# ping 2001:db8:1::2
```

Remember

Both ends need compatible IPv6 addressing and a valid route. In some lab environments, IPv6 forwarding must be enabled explicitly.

Quick Recap

- **Forwarding**: selects an outgoing interface/next hop for an individual packet.
- **Static route**: manually configured; **default route**: fallback for unmatched destinations.
- **Distance vector**: learns distances from neighbours; RIP uses hop count.
- **Link state**: builds topology database and runs SPF; OSPF uses Dijkstra.
- **Path vector**: advertises AS paths and supports policy decisions; BGP is the key example.
- **EIGRP**: advanced distance vector using DUAL.
- **Multicast**: one sender to a group; IPv4 Class D range is 224.0.0.0/4.
- **IPv6**: 128-bit addresses written in hexadecimal.

Questions

1. What is the difference between routing and forwarding?
2. What is a default route, and when is it used?
3. A router has a direct route to a destination at cost 7 and a route through a neighbour with costs 1 and 2. Which path is preferred by a distance-vector calculation?
4. Why does RIP treat 16 hops as unreachable?
5. Which algorithm does OSPF use to calculate shortest paths?
6. What information in BGP helps prevent AS-path loops?
7. What is the difference between unicast and multicast?
8. Compress `2001:0db8:0000:0000:0000:0000:0000:0001`.

Exam-Oriented Questions

Short-answer questions

- Differentiate RIP v1 and RIP v2.
- Define LSDB, LSA and OSPF Area 0.
- Explain successor and feasible successor in EIGRP.
- State the purpose of an AS-PATH attribute in BGP.

Long-answer / diagram questions

- Explain distance-vector routing with a worked example.
- Explain Dijkstra's algorithm using a weighted network graph.
- Compare RIP, OSPF, EIGRP and BGP.
- Explain IPv6 notation, address types and prefix length with examples.

Experiment

Activity: Choose the Route

Draw four routers with at least two possible paths between a source and destination. Assign a cost to every link.

1. Calculate the total cost of each path.
2. Identify the least-cost path.
3. Explain what a distance-vector router would learn from a neighbour.
4. Explain how a link-state router would calculate its shortest-path tree.
5. Discuss why BGP might select a route for policy reasons instead of the numerically shortest route.

References and Further Reading

- Kurose and Ross, *Computer Networking: A Top-Down Approach*.
- Andrew S. Tanenbaum and David J. Wetherall, *Computer Networks*.
- RFC 1058: Routing Information Protocol.
- RFC 2453: RIP Version 2.
- RFC 2328: OSPF Version 2.
- RFC 4271: A Border Gateway Protocol 4 (BGP-4).
- RFC 8200: Internet Protocol, Version 6 (IPv6) Specification.